KDDM 2 Challenge - Anomaly detection in time series data

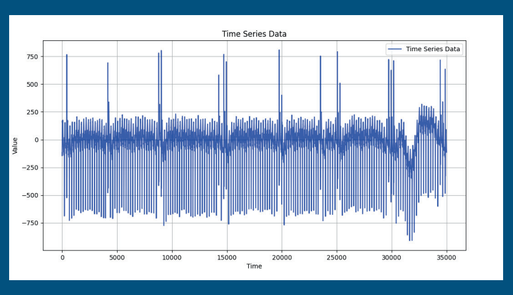
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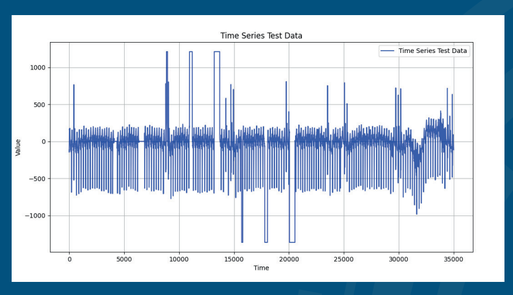
Challenge

Develop a method for **anomaly detection in time series data**
Given:

- Time series dataset with 3500 “normal“ samples

Task: Identify and classify **collective anomalies** within a test set that adheres to the same data distribution as the trainig data





Approach

What anomalies are there?

- Constant, Peak, Through, Reverse sequence, Noise
  - Create testfile

What algorithm to use?

- Tried LSTM and KNN → KNN more promising
- Grid search
- Can we improve?
  - smoothing with gaussian kernel

Window size	K	Threshold factor	Distance metric
50	3	1.5	euclidean
100	5	3.5	manhattan
250	10	5	cosine

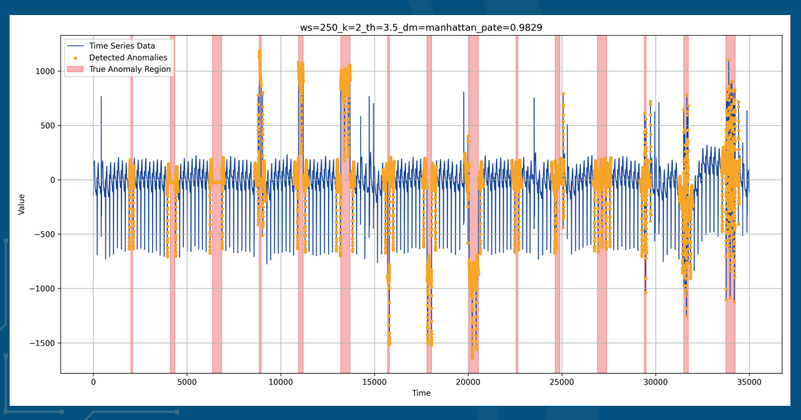
Initially, the grid search on unprocessed data achieved a PATE score of **0.714**. By applying a Gaussian filter to smooth the anomaly scores, we improved the PATE score to **0.983**

Final Solution

Our final model uses the following parameters and applies Gaussian smoothing to the anomaly scores with a sigma of 75:

Window size	K	Threshold factor	Distance metric
250	2	3.5	Manhattan

- Final PATE score on test data: 0.983
- The model significantly outperformed the baseline on the test set



Evaluation

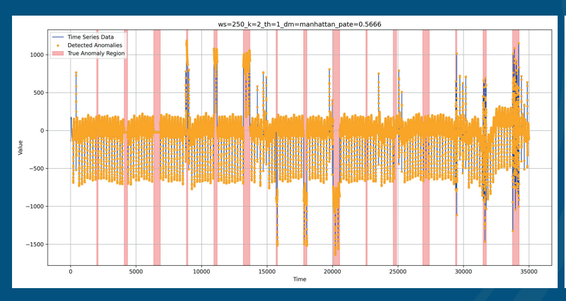
Baseline

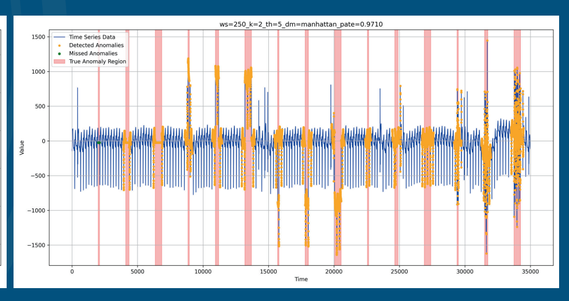
- Label everything as anomaly: PATE score: 0.57
- Model config found with Grid search: PATE score: 0.71

Sensitivity Analysis

We tested the impact of key parameters on the smoothed data:

- Window Size: Minimal impact
  - Tested sizes: 50, 100, 400
  - PATE scores ranged from 0.85 to 0.87
- Threshold Factor: High impact
  - A low threshold overclassifies data as anomalies, while a threshold factor of 5 achieves near-perfect performance on the test data





- K (Number of Neighbors): Significant impact
  - Tested values: 1, 5, 10
  - PATE score decreased from 0.97 (K=1) to 0.65 (K=10)

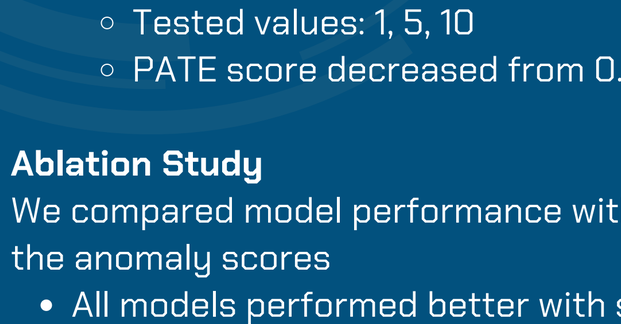
Ablation Study

We compared model performance with and without smoothing the anomaly scores

- All models performed better with smoothing

Error Study

- **Higher K values** resulted in more errors due to overgeneralization
- **Shorter Sequences:** constants, noise
- **Longer Sequences:** constants, reverse, noise
- **Data Stationarity**: Differencing data reduced the model's ability to detect anomalies effectively
  - **Problem:** loss of information



Lessons Learned

- **V**isualization more important than metric
- Differencing data comes with loss of information
- Don’t rush - take time to understand data
- Invest more time in preprocessing